

2026-05-07-p2-f26-openhands-workspace-design

Phase 2 / Feature 26 — Openhands Workspace + Task Planner + Step Executor

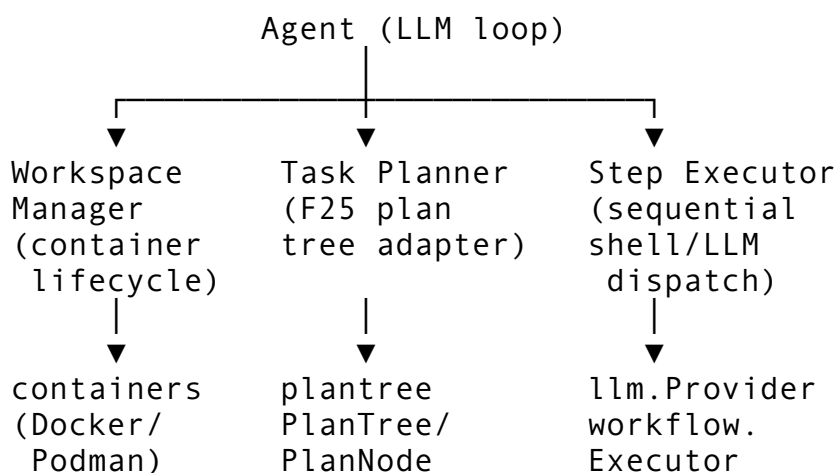
Date: 2026-05-07 **Status:** Approved (auto-approved per programme cadence) **Programme:** CLI-Agent Fusion — Phase 2 port (openhands — workspace, planner, executor)

Programme position: F26 is the **sixth** Phase 2 feature (F21-F25 shipped before it).
T01 appends an F26 evidence section to `docs/improvements/07_phase_2_evidence.md`.

1. Goal

Ship a real, end-to-end **workspace + task planner + step executor** subsystem for the HelixCode CLI agent, modelled on Openhands' agent orchestration architecture (`cli_agents/` `openhands/`). Three components: (1) container-based per-task workspaces managed through the containers submodule (`digital.vasic.containers`), (2) a sequential step executor that walks F25 plan trees and dispatches shell/LLM steps, and (3) a task planner that bridges F25's structured plan trees with Openhands-style execution pipelines. Tools surface: `workspace_create`, `workspace_list`, `workspace_cleanup`, `task_plan`, `task_step`. Slash: `/openhands`. Cobra: `helixcode workspace {create,list,cleanup}`.

2. Architecture



2.1 Workspace Manager

- **Package:** `internal/workspace/` (NEW — avoids collision with `internal/tools/worktree/` from F04)

- Container-based per-task workspaces using `digital.vasic.containers` orchestrator
- Lifecycle: `CreateWorkspace` → `RunTask` → `CleanupWorkspace`
- Each workspace is a Docker/Podman container with:
 - Mounted project directory (read-write)
 - Isolated network (default deny; opt-in for web tasks)
 - Resource limits (CPU/memory caps)
 - Auto-cleanup TTL (idle timeout)

2.2 Task Planner

- **Package:** `internal/planner/` (NEW)
- Bridges F25 plan trees with execution: converts a `PlanNode` to a `TaskStep` chain
- Validates dependencies between steps
- Tracks execution state (`Pending` → `Running` → `Completed/Failed`)
- Reports progress back to the plan tree (updates `PlanNode.Status`)

2.3 Step Executor

- **Package:** `internal/planner/executor.go` (within planner package)
- Sequential step execution with retry
- Two executor types: `ShellStep` (real shell execution via F06 executor) and `LLMStep` (LLM provider call)
- Each step has: timeout, retry count, action type, parameters
- Status propagation to plan tree nodes

3. Data Model

3.1 Workspace

```
type Workspace struct {
    ID          string    // UUID
    Name        string    // human-readable
    ContainerID string    // Docker/Podman container ID
    Image       string    // container image (default:
                alpine:latest)
    ProjectDir  string    // host path mounted into container
    Status      WorkspaceStatus // Creating/Running/Stopped/
                Error
    CreatedAt   time.Time
    TTL         time.Duration // idle timeout before auto-
                cleanup
}
```

3.2 TaskStep

```
type StepType int
const (
    StepShell StepType = iota
```

```

        StepLLM
    )

type TaskStep struct {
    ID            string
    PlanNodeID    string           // backlink to F25 PlanNode
    Type          StepType
    Command       string           // for StepShell
    Prompt       string           // for StepLLM
    Timeout       time.Duration
    MaxRetries    int
    Status        StepStatus      // Pending/Running/Completed/
                        Failed
    Output        string
    Error         string
    StartedAt     time.Time
    CompletedAt   time.Time
}

```

3.3 TaskPlan

```

type TaskPlan struct {
    ID            string
    Name          string
    WorkspaceID   string
    Steps         []TaskStep
    Status        PlanStatus
    PlanTree      *plantree.PlanTree // backlink to F25
}

```

4. Components

4.1 New files

File	Purpose
internal/workspace/ types.go	Workspace, WorkspaceStatus, sentinels
internal/workspace/ manager.go	Container lifecycle via containers orchestrator
internal/workspace/ workspace_tools.go	workspace_create/list/cleanup tools.Tool impls
internal/planner/ types.go	TaskStep, TaskPlan, StepType, StepStatus
internal/planner/ executor.go	Sequential step executor with retry

File	Purpose
internal/planner/ planner_tools.go	task_plan, task_step tools.Tool impls
internal/commands/ openhands_command.go	/openhands slash
cmd/cli/workspace_cmd.go	Cobra subcommands

4.2 Modified files

File	Change
cmd/cli/main.go	Wire workspace manager + planner + tools + slash
internal/tools/registry.go	Register CategoryWorkspace
internal/commands/builtin/ register.go	Register /openhands

5. Anti-Bluff Hot Zone

Five critical degenerate patterns:

1. **Workspace created but container not running** — `docker ps` after `CreateWorkspace` MUST show container
2. **Step executor claims completed but command never ran** — `step.Output` MUST contain real command output
3. **Task plan references workspace that doesn't exist** — Validate workspace ID before starting plan
4. **Cleanup claims success but container still running** — `docker ps` after `CleanupWorkspace` MUST NOT show container
5. **Executor returns success on failed step** — `StepStatus` MUST match exit code / LLM error

6. Task Breakdown

#	Task	Description
T01	Bootstrap	F26 evidence + advance PROGRESS + CONTINUATION
T02	workspace/types.go + manager.go	Workspace + Manager (container lifecycle) (TDD)
T03	workspace/ workspace_tools.go	workspace_create/list/cleanup tools (TDD)
T04	planner/types.go + executor.go	TaskStep + TaskPlan + StepExecutor (TDD)
T05	planner/planner_tools.go	task_plan, task_step tools (TDD)

#	Task	Description
T06	/openhands slash + Cobra subcommands	(TDD)
T07	main.go wiring	Register tools + slash + cobra
T08	Challenge harness + close-out	5 phases + push 4 remotes

F26 spec — Openhands workspace + task planner + step executor.